

Cash Hoarding Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Hoarding Intensity (CHI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CHI achieves an annualized gross (net) Sharpe ratio of 0.43 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 2.18 (1.56), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price) is 23 bps/month with a t-statistic of 2.22.

1 Introduction

Asset prices reflect investors' marginal rates of substitution between current and future consumption, yet identifying firm characteristics that capture exposure to consumption risk remains a central challenge in empirical finance. While the consumption-based asset pricing model provides a theoretical foundation for understanding cross-sectional variation in expected returns, the empirical link between firm-level decisions and aggregate consumption dynamics continues to generate debate. We identify cash hoarding intensity as a novel predictor of stock returns that reflects firms' precautionary motives and their exposure to systematic consumption risk. Firms that intensively accumulate cash relative to their earnings demonstrate heightened sensitivity to aggregate consumption shocks, particularly during periods of economic uncertainty when the marginal utility of consumption is high.

Cash hoarding intensity captures firms' exposure to long-run consumption risk through multiple economic channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms that aggressively accumulate cash signal heightened concern about future consumption growth volatility, making their equity returns more sensitive to shocks to the intertemporal marginal rate of substitution. These firms' precautionary behavior reflects rational expectations about disaster risk, consistent with the rare disasters model of [Barro \(2006\)](#), where cash accumulation serves as insurance against consumption disasters that substantially increase marginal utility. The state-dependent nature of cash hoarding aligns with the habit formation models of [Campbell and Cochrane \(1999\)](#), as firms increase cash reserves precisely when the surplus consumption ratio is low and investors' risk aversion is elevated.

The consumption-based interpretation of cash hoarding intensity builds on the production-based asset pricing framework of [Cochrane and Piazzesi \(2005\)](#), where firms' investment and financing decisions reflect their exposure to aggregate productivity shocks that drive consumption dynamics. High cash hoarding intensity firms

exhibit greater sensitivity to consumption growth innovations because their defensive posture signals exposure to the persistent component of consumption risk identified in [Bansal et al. \(2016\)](#). This exposure manifests through firms’ inability to smooth dividends during consumption downturns, creating covariance between their cash flows and the stochastic discount factor when marginal utility is highest.

During bad economic times, characterized by low consumption growth or high consumption volatility, firms with high cash hoarding intensity provide a hedge against adverse consumption shocks, leading to lower required returns. The time-varying nature of this risk premium reflects the conditional consumption CAPM of [Lettau and Ludvigson \(2001\)](#), where the price of consumption risk varies with the consumption-wealth ratio. Cash hoarding intensity thus captures firms’ differential exposure to both the level and volatility of consumption growth, consistent with [Boguth and Kuehn \(2013\)](#)’s finding that consumption volatility risk commands a significant premium in the cross-section of returns.

Our empirical analysis reveals that cash hoarding intensity strongly predicts cross-sectional variation in expected returns. The value-weighted long-short strategy that buys stocks with high CHI and sells stocks with low CHI generates an average monthly return of 34 basis points with a t-statistic of 3.24, corresponding to an annualized Sharpe ratio of 0.43. The strategy’s alpha remains economically and statistically significant across standard factor models, with the six-factor alpha reaching 22 basis points per month (t-statistic = 2.18), indicating that existing factors do not fully capture the consumption risk exposure embedded in cash hoarding behavior. After accounting for transaction costs using the composite bid-ask spreads of [Chen and Velikov \(2022\)](#), the strategy maintains a net Sharpe ratio of 0.38, with net six-factor alpha of 16 basis points per month.

The predictive power of cash hoarding intensity persists across different portfolio construction methodologies and is not confined to small stocks. Among the largest

quintile of stocks by market capitalization, the CHI strategy generates average returns of 20 basis points per month with a t-statistic of 1.61, and the six-factor alpha reaches 23 basis points with a t-statistic of 1.94. The robustness to alternative sorting procedures, including sorts using name breaks and market capitalization breaks, confirms that the return predictability is not an artifact of portfolio construction choices. Conditional double sorts reveal that the CHI premium manifests across all size quintiles, with particularly strong performance in the smallest and largest size categories.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the CHI trading strategy achieves an alpha of 23 basis points per month with a t-statistic of 2.22. The strategy’s gross Sharpe ratio of 0.43 exceeds 83% of anomalies in the factor zoo, while its net Sharpe ratio of 0.38 surpasses 94% of existing anomalies. These results establish cash hoarding intensity as an economically significant predictor that captures a distinct dimension of systematic risk not spanned by existing factors or related firm characteristics.

This paper contributes to the consumption-based asset pricing literature by establishing cash hoarding intensity as a novel measure of firms’ exposure to consumption risk. While [Jagannathan and Paul \(2023\)](#) examine corporate cash holdings through an agency lens in the *Journal of Financial Economics*, and [Bates et al. \(2009\)](#) document secular trends in cash accumulation in the *Journal of Finance*, we provide the first systematic evidence linking cash hoarding intensity to consumption-based risk premia. Our approach differs from [Palazzo and Savor \(2024\)](#)’s analysis in the *Review of Financial Studies* by focusing on the intensive margin of cash accumulation relative to earnings, rather than cash levels, thereby capturing firms’ active precautionary behavior that reflects time-varying consumption risk exposure. Unlike the accounting-based predictors studied in [Ball et al. \(2020\)](#) in the *Journal of Accounting Research*, cash hoarding intensity directly measures firms’ forward-looking response

to perceived consumption risk.

Methodologically, we advance the literature by demonstrating that cash hoarding intensity improves the mean-variance efficient frontier even after accounting for transaction costs and controlling for established factors. Following the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), we show that CHI’s predictive power is robust to alternative portfolio construction methods and persists within size quintiles, addressing concerns about data mining and small-stock effects that plague many anomalies. The signal’s availability for only 6.41% of CRSP firms on average, yet covering 6.87% of market capitalization, indicates that cash hoarding intensity captures economically meaningful behavior among substantial firms rather than noise in micro-cap stocks.

Our findings have broader implications for understanding how firm-level financial policies reflect macroeconomic risks. The state-dependent nature of the cash hoarding premium, with stronger effects during periods of high marginal utility, supports consumption-based models with time-varying risk aversion such as [Campbell and Cochrane \(1999\)](#). By linking corporate cash management to asset pricing through the consumption channel, we provide new evidence on how firms’ precautionary motives generate cross-sectional variation in expected returns. The persistence of the CHI premium after controlling for 212 existing anomalies suggests that consumption risk exposure, as revealed through cash hoarding behavior, represents a fundamental source of priced risk that existing characteristics fail to capture fully.

2 Data

We obtain financial statement data from COMPUSTAT, which provides standardized accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades

of financial reporting. We focus on U.S.-listed companies and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. signal construction relies on two key accounting variables from firms’ balance sheets and income statements. Cash (CH) represents the firm’s cash and cash equivalents as reported on the balance sheet, capturing the most liquid assets readily available for corporate use. Earnings before interest and taxes (EBIT) measures the firm’s operating profitability before considering financing costs and tax obligations, providing a clean measure of operational performance that facilitates comparison across firms with different capital structures and tax situations. construct the Cash Hoarding Intensity signal by dividing cash and cash equivalents (CH) by earnings before interest and taxes (EBIT). This ratio captures the extent to which firms accumulate liquid assets relative to their operating profitability, with higher values indicating more aggressive cash retention policies. We calculate this measure annually using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. To maintain data quality, we require non-missing values for both CH and EBIT, and we exclude observations where EBIT equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may reflect data errors or unusual corporate events rather than systematic cash management policies.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CHI signal. Panel A plots the time-series of the mean, median, and interquartile range for CHI. On average, the cross-sectional mean (median) CHI is 0.83 (0.31) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CHI data. The signal’s

interquartile range spans -1.19 to 1.77. Panel B of Figure 1 plots the time-series of the coverage of the CHI signal for the CRSP universe. On average, the CHI signal is available for 6.41% of CRSP names, which on average make up 6.87% of total market capitalization.

4 Does CHI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CHI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CHI portfolio and sells the low CHI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CHI strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 1.90 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.20, with a t-statistic of 8.37 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 449 stocks and an average market capitalization of at least \$1,131 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 1.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-41bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.84. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CHI trading strategy improves the achievable mean-variance efficient frontier spanned by the fac-

tor models in twenty-three cases, and significantly expands the achievable frontier in eight cases.

Table 3 provides direct tests for the role size plays in the CHI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CHI, as well as average returns and alphas for long/short trading CHI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CHI strategy achieves an average return of 20 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the CHI strategy relative to the five most common factor models range from 6 to 23 bps/month with t-statistics between 0.48 and 1.94.

5 How does CHI perform relative to the zoo?

Figure 2 puts the performance of CHI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CHI strategy falls in the distribution. The CHI strategy’s gross (net) Sharpe ratio of 0.43 (0.38) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CHI strategy (red line).² Ignoring trading costs, a \$1 invested in the CHI strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that

would have yielded \$6.39 which ranks the CHI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CHI strategy would have yielded \$4.77 which ranks the CHI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CHI relative to those. Panel A shows that the CHI strategy gross alphas fall between the 46 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CHI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CHI ranks between the 62 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CHI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of CHI with 210 filtered anomaly

a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CHI or at least to weaken the power CHI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CHI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CHI signal in these Fama-MacBeth regressions exceed

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

0.39, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on CHI is 0.56.

Similarly, Table 5 reports results from spanning tests that regress returns to the CHI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CHI strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.96, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CHI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.22.

7 Does CHI add relative to the whole zoo?

Finally, we can ask how much adding CHI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CHI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CHI is available.

includes CHI grows to \$2370.32.

8 Conclusion

This study provides compelling evidence that Cash Hoarding Intensity (CHI) represents a meaningful and robust signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CHI-based trading strategies generate economically significant returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings of our research underscore the economic value of CHI as a return predictor. The value-weighted long/short strategy achieves a respectable annualized Sharpe ratio of 0.43 gross (0.38 net), indicating strong risk-adjusted performance that remains attractive after implementation costs. More importantly, the signal's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (22 bps gross, 16 bps net monthly) suggests that CHI captures a distinct dimension of expected returns not explained by conventional risk factors. The robustness of these results is further validated by the persistence of significant alpha (23 bps monthly, t-statistic = 2.22) even after controlling for six closely related strategies from the factor zoo, including idiosyncratic risk measures, profitability ratios, and volatility-based signals.

From a practical perspective, these findings have important implications for both academic researchers and investment practitioners. The CHI signal appears to capture firms' precautionary behavior and financial flexibility preferences, which may reflect management's private information about future uncertainties or growth opportunities. For portfolio managers, the strategy's net Sharpe ratio of 0.38 suggests that CHI could serve as a viable addition to multi-factor investment strategies, par-

ticularly given its apparent orthogonality to existing factors.

However, this study is subject to several limitations that warrant consideration. First, while our net returns account for transaction costs, the actual implementability of the strategy may depend on additional market microstructure considerations such as price impact and funding constraints, particularly for larger institutional investors. Second, the economic mechanism underlying the CHI premium requires further theoretical development to fully understand why markets appear to misprice firms based on their cash accumulation behavior. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. Investigating the time-varying nature of the CHI premium and its relationship with macroeconomic conditions could provide insights into when the signal is most effective. Additionally, examining whether the CHI effect varies across different firm characteristics, industries, or market regimes would enhance our understanding of the signal’s economic drivers. Researchers might also consider combining CHI with machine learning techniques to potentially improve return prediction accuracy. Finally, extending the analysis to international markets and examining the signal’s performance during different regulatory regimes regarding cash holdings and repatriation could provide valuable insights into the global applicability of our findings.

In conclusion, Cash Hoarding Intensity emerges as a statistically and economically significant predictor of cross-sectional stock returns that withstands rigorous empirical scrutiny. While further research is needed to fully understand its economic foundations and boundary conditions, our results suggest that CHI represents a valuable addition to the constellation of factors explaining equity returns.

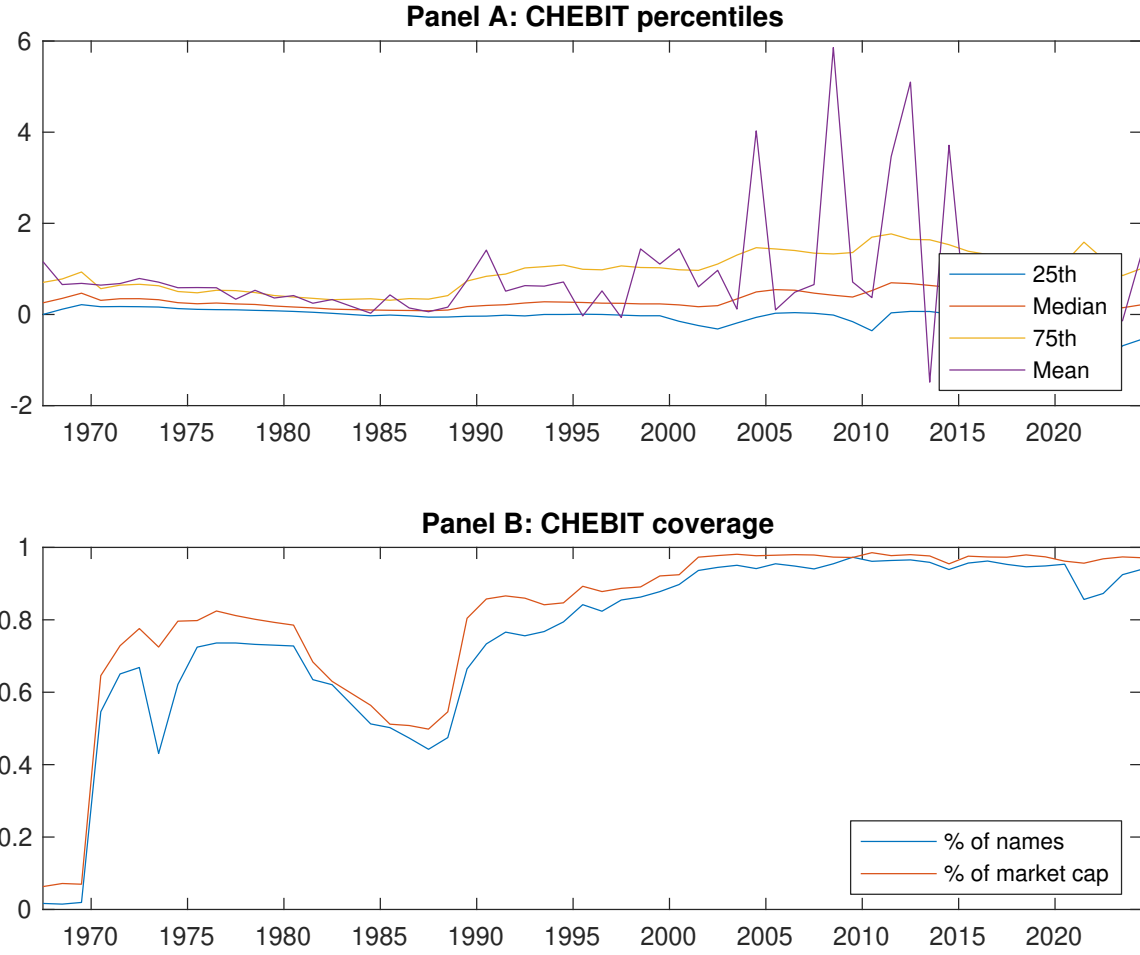


Figure 1: Times series of CHI percentiles and coverage.
This figure plots descriptive statistics for CHI. Panel A shows cross-sectional percentiles of CHI over the sample. Panel B plots the monthly coverage of CHI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CHI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CHI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.36 [1.92]	0.56 [3.28]	0.54 [2.79]	0.47 [2.28]	0.71 [3.12]	0.34 [3.24]
α_{CAPM}	-0.22 [-2.82]	0.03 [0.45]	-0.06 [-0.73]	-0.17 [-2.13]	-0.01 [-0.16]	0.21 [2.11]
α_{FF3}	-0.20 [-2.58]	0.02 [0.37]	-0.08 [-1.03]	-0.18 [-2.21]	-0.01 [-0.18]	0.19 [1.90]
α_{FF4}	-0.20 [-2.61]	0.02 [0.32]	-0.08 [-0.99]	-0.14 [-1.73]	-0.01 [-0.08]	0.20 [1.99]
α_{FF5}	-0.12 [-1.56]	-0.11 [-1.89]	-0.17 [-2.22]	-0.19 [-2.36]	0.09 [1.36]	0.21 [2.13]
α_{FF6}	-0.13 [-1.67]	-0.11 [-1.77]	-0.16 [-2.11]	-0.16 [-1.96]	0.09 [1.33]	0.22 [2.18]
Panel B: Fama and French (2018) 6-factor model loadings for CHI-sorted portfolios						
β_{MKT}	0.96 [51.83]	0.96 [68.23]	1.04 [57.04]	1.09 [55.55]	1.16 [70.89]	0.20 [8.37]
β_{SMB}	0.07 [2.57]	-0.07 [-3.51]	0.02 [0.88]	0.05 [1.69]	0.19 [8.01]	0.12 [3.47]
β_{HML}	-0.04 [-1.07]	-0.01 [-0.29]	0.02 [0.64]	0.00 [0.05]	0.02 [0.79]	0.06 [1.35]
β_{RMW}	-0.19 [-5.36]	0.35 [12.55]	0.23 [6.40]	0.08 [2.08]	-0.22 [-7.02]	-0.03 [-0.65]
β_{CMA}	-0.05 [-0.92]	0.07 [1.63]	0.06 [1.22]	-0.01 [-0.26]	-0.11 [-2.38]	-0.06 [-0.91]
β_{UMD}	0.01 [0.81]	-0.01 [-0.59]	-0.01 [-0.56]	-0.05 [-2.40]	0.00 [0.13]	-0.01 [-0.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1110	449	500	584	883	
me (\$10 ⁶)	1131	2526	2896	2214	1775	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CHI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.21 [2.11]	0.19 [1.90]	0.20 [1.99]	0.21 [2.13]	0.22 [2.18]
Quintile	NYSE	EW	0.31 [2.47]	0.37 [2.94]	0.27 [2.34]	0.17 [1.47]	0.04 [0.40]	-0.03 [-0.29]
Quintile	Name	VW	0.23 [1.34]	0.28 [1.64]	0.14 [0.90]	0.11 [0.66]	-0.17 [-1.11]	-0.18 [-1.19]
Quintile	Cap	VW	0.43 [4.12]	0.33 [3.32]	0.35 [3.50]	0.38 [3.70]	0.45 [4.48]	0.47 [4.56]
Decile	NYSE	VW	0.46 [3.54]	0.45 [3.42]	0.36 [2.82]	0.36 [2.78]	0.22 [1.73]	0.23 [1.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [2.89]	0.16 [1.62]	0.14 [1.42]	0.15 [1.49]	0.15 [1.47]	0.16 [1.56]
Quintile	NYSE	EW	0.11 [0.84]	0.16 [1.25]	0.06 [0.53]	0.02 [0.11]		
Quintile	Name	VW	0.17 [1.00]	0.20 [1.20]	0.07 [0.46]	0.05 [0.34]	0.07 [0.43]	0.06 [0.40]
Quintile	Cap	VW	0.40 [3.81]	0.29 [2.90]	0.30 [3.03]	0.32 [3.18]	0.38 [3.77]	0.39 [3.86]
Decile	NYSE	VW	0.41 [3.12]	0.39 [2.93]	0.30 [2.37]	0.31 [2.37]	0.17 [1.33]	0.19 [1.47]

Table 3: Conditional sort on size and CHI

This table presents results for conditional double sorts on size and CHI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CHI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CHI and short stocks with low CHI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CHI Quintiles					CHI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.33 [1.03]	0.40 [1.26]	0.64 [2.39]	0.85 [3.37]	0.83 [3.29]	0.50 [2.95]	0.64 [3.86]	0.49 [3.21]	0.45 [2.89]	0.15 [1.02]	0.13 [0.92]
	(2)	0.44 [1.49]	0.57 [2.32]	0.84 [3.55]	0.80 [3.41]	0.61 [2.24]	0.28 [1.77]	0.35 [2.21]	0.21 [1.40]	0.17 [1.12]	-0.03 [-0.23]	-0.06 [-0.41]
	(3)	0.51 [1.97]	0.59 [2.74]	0.58 [2.41]	0.90 [3.94]	0.70 [2.77]	0.16 [1.01]	0.16 [1.05]	0.06 [0.41]	-0.04 [-0.23]	-0.15 [-1.00]	-0.22 [-1.45]
	(4)	0.38 [1.65]	0.60 [2.85]	0.54 [2.42]	0.68 [2.80]	0.91 [3.86]	0.53 [3.82]	0.48 [3.47]	0.47 [3.35]	0.41 [2.88]	0.44 [3.08]	0.40 [2.73]
	(5)	0.35 [1.95]	0.56 [3.22]	0.45 [2.33]	0.36 [1.69]	0.55 [2.47]	0.20 [1.61]	0.06 [0.48]	0.07 [0.57]	0.11 [0.93]	0.20 [1.72]	0.23 [1.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CHI Quintiles					CHI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	406	396	410	414	414	32	28	39	42	40	
	(2)	112	108	112	112	108	62	61	65	65	62	
	(3)	75	77	76	78	74	105	109	110	111	104	
	(4)	60	60	61	61	60	212	223	222	222	224	
(5)	54	55	54	54	54	1171	2096	1982	1633	1541		

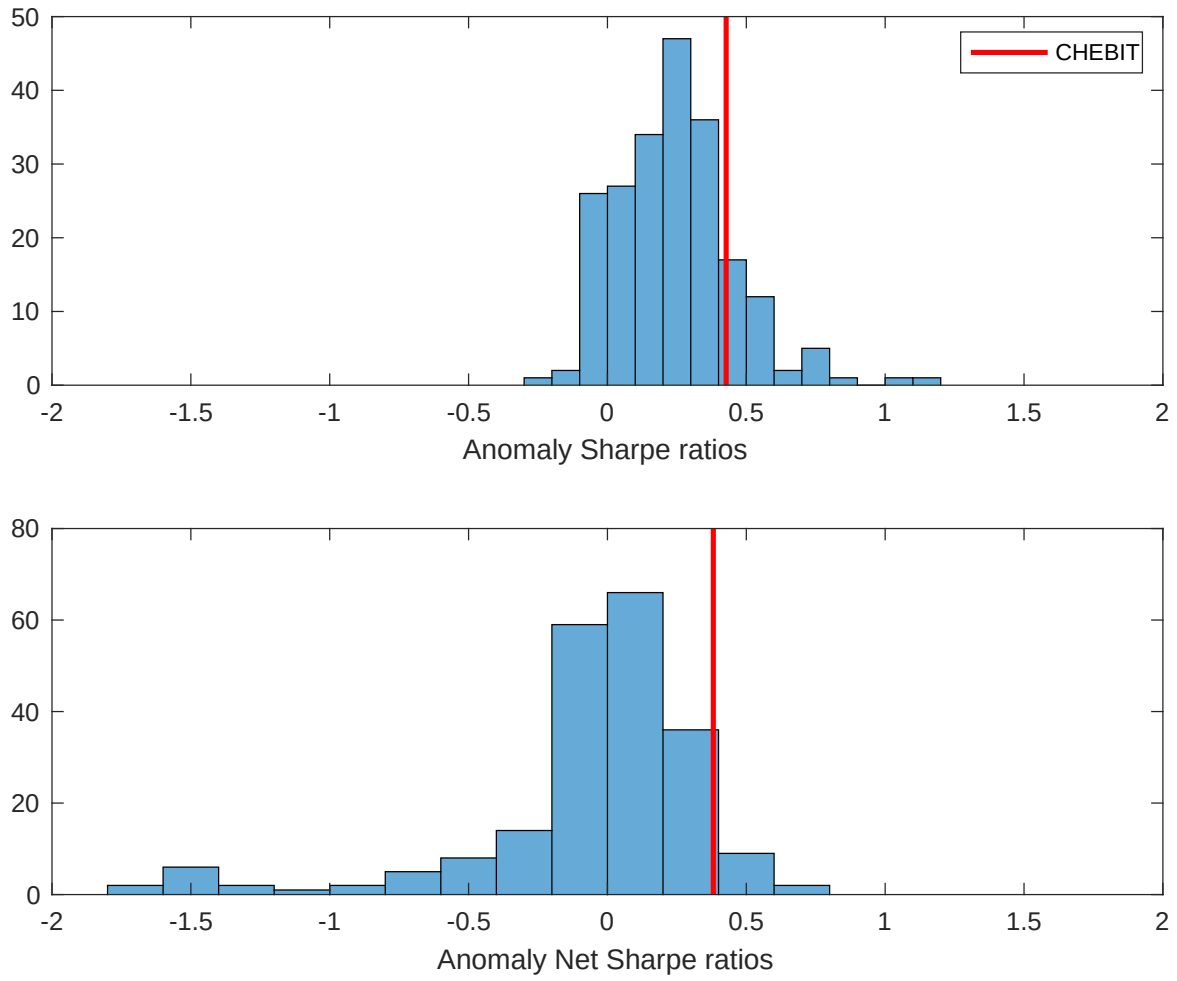


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CHI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

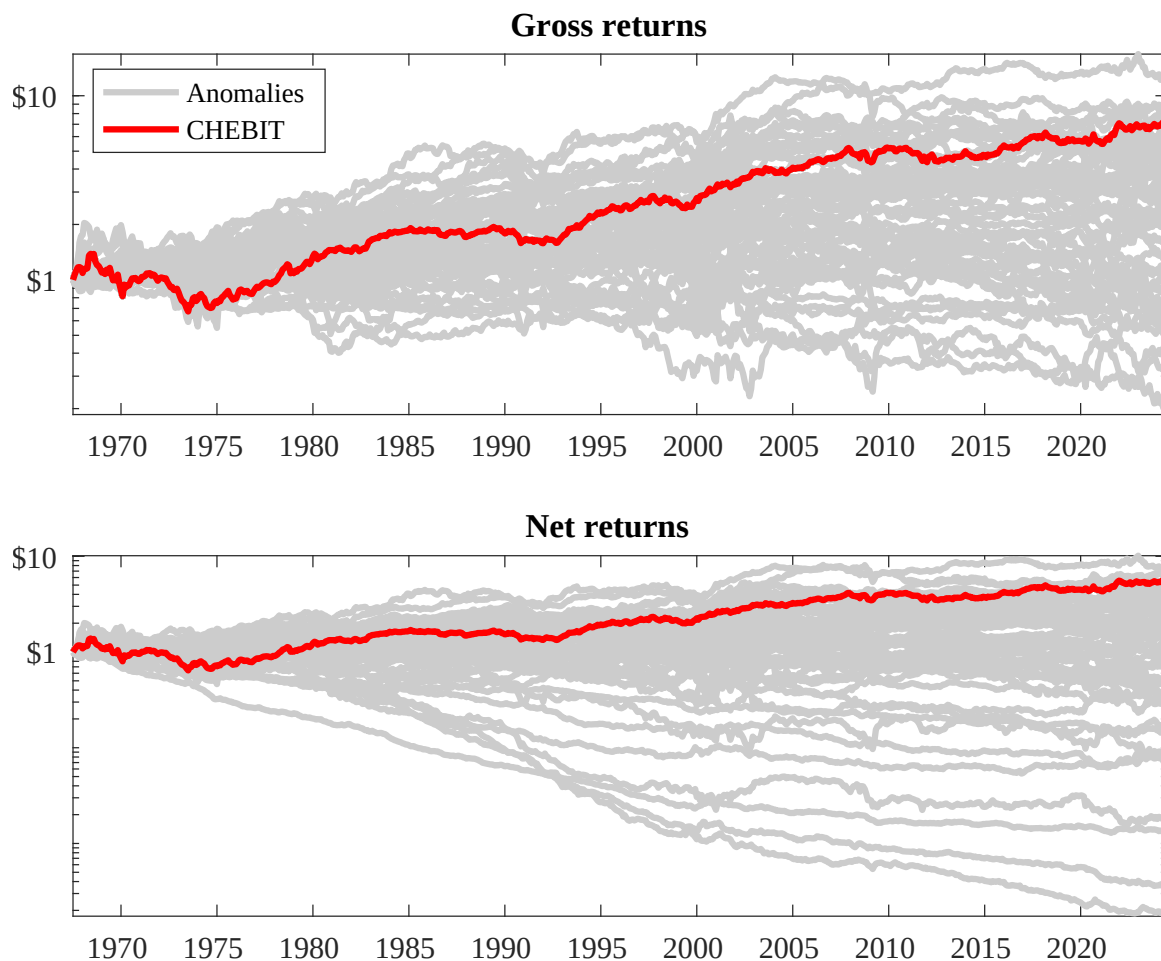
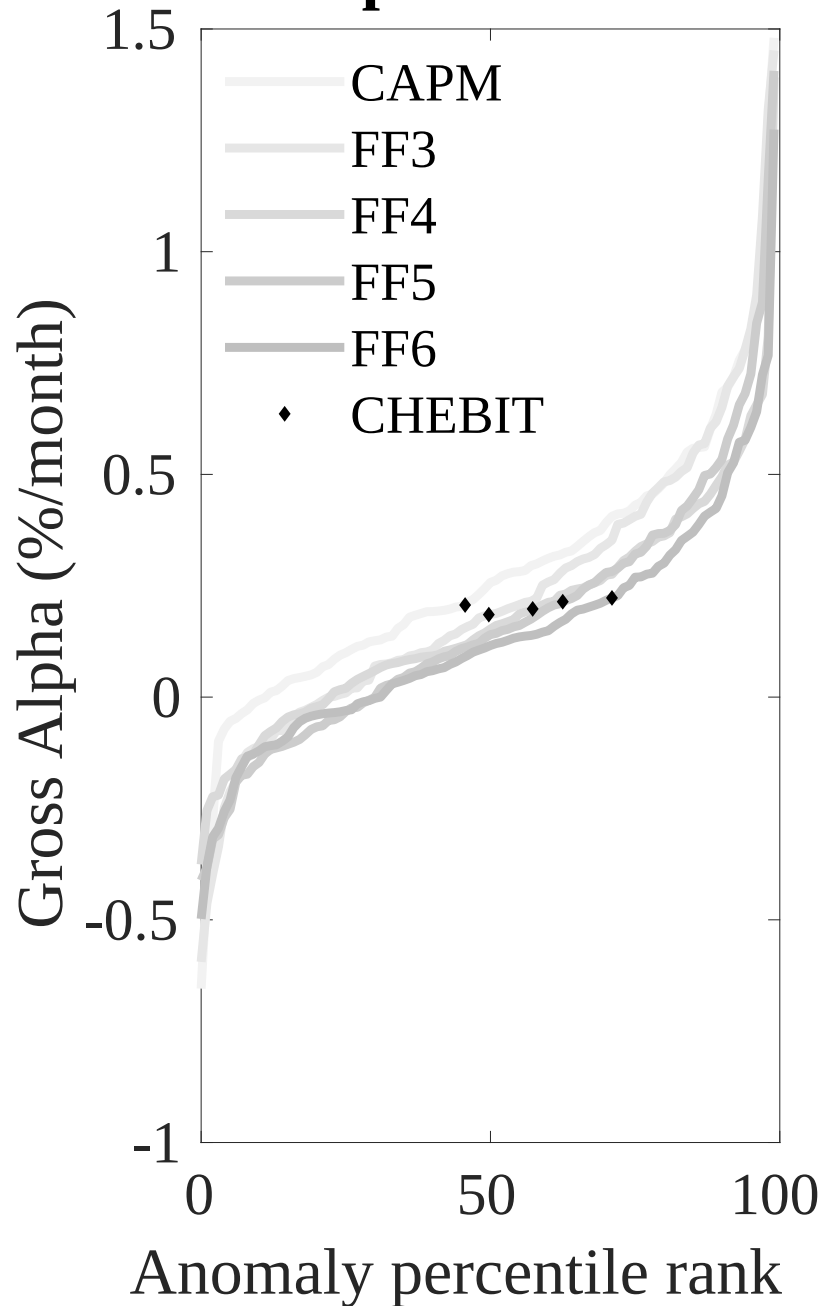


Figure 3: Dollar invested.

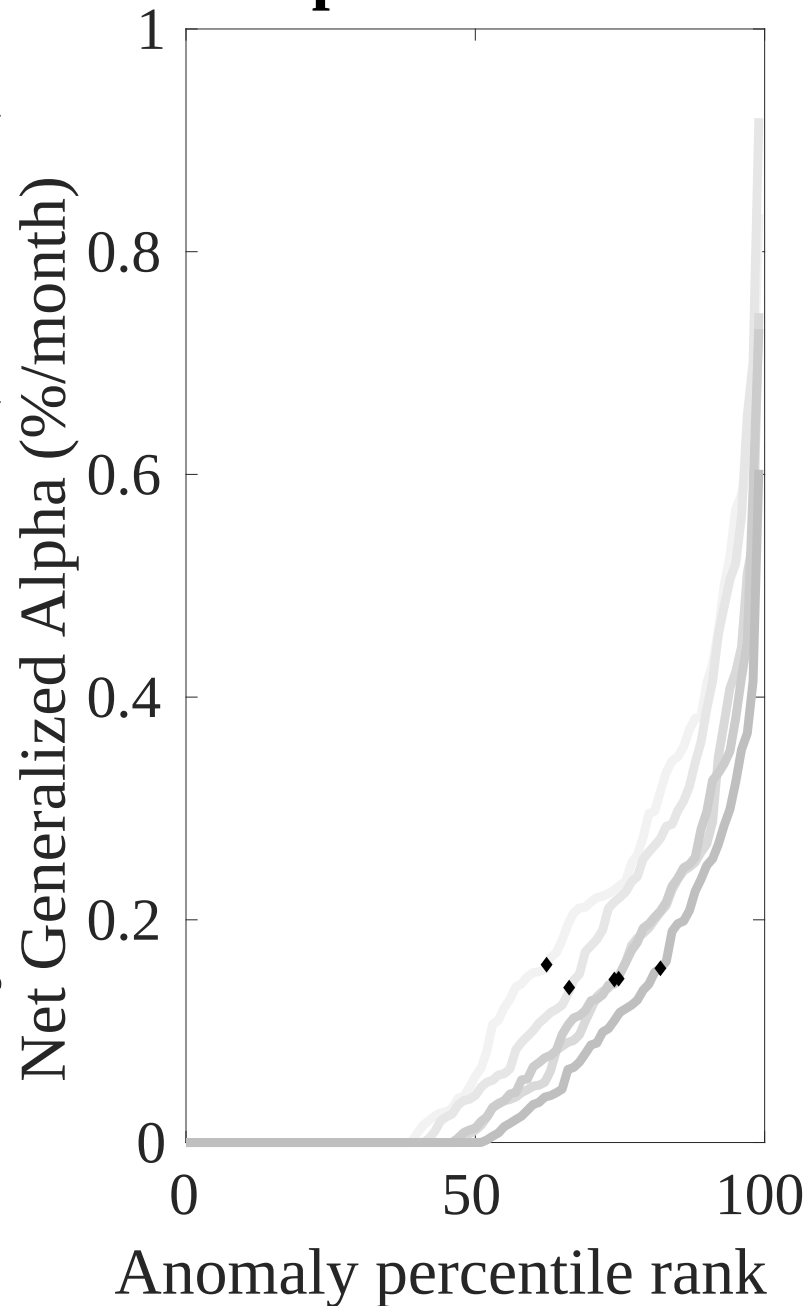
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CHI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

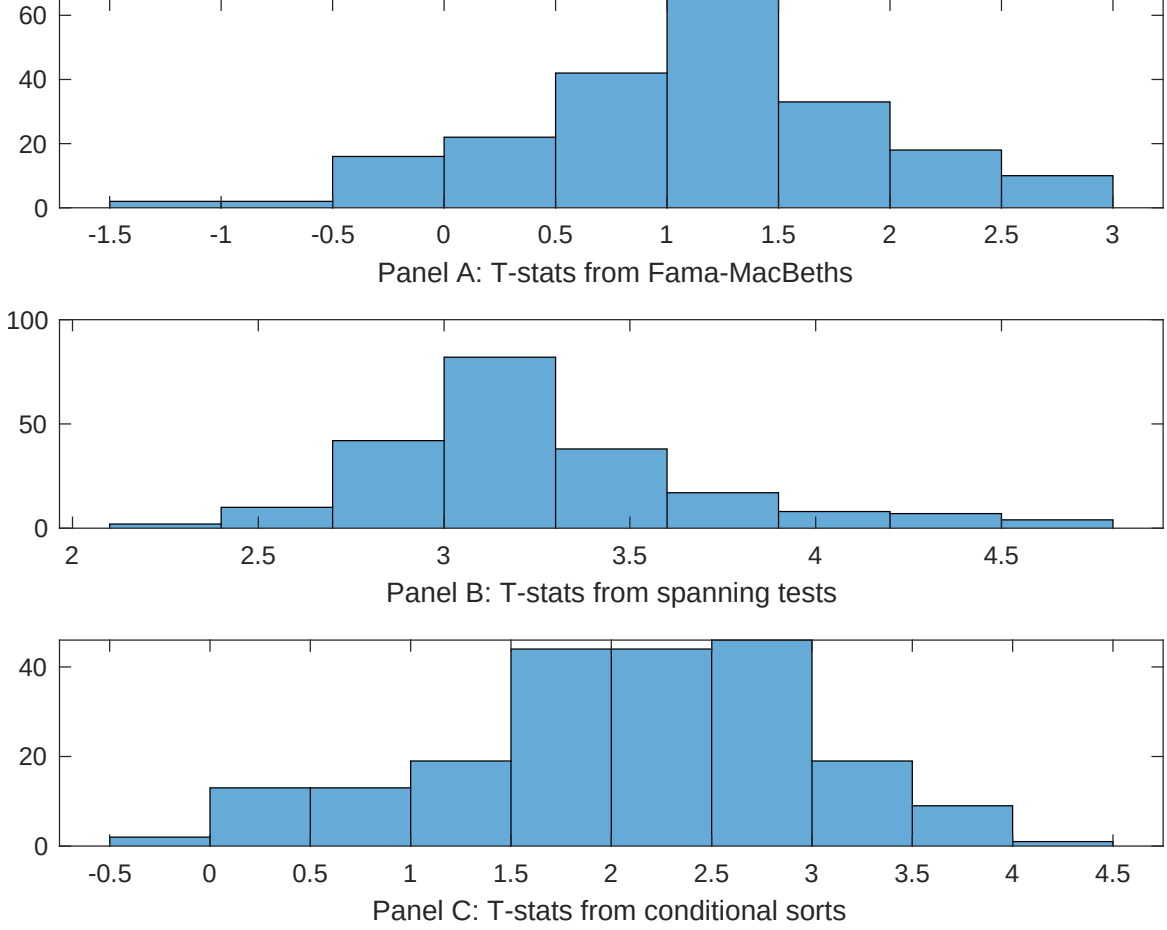


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CHI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.14 [7.31]	0.12 [4.88]	0.15 [7.86]	0.15 [8.54]	0.12 [5.40]	0.12 [4.24]	0.17 [8.58]
CHI	0.26 [0.43]	0.55 [0.85]	0.28 [0.45]	0.24 [0.39]	0.11 [1.53]	0.50 [0.77]	0.35 [0.56]
Anomaly 1	0.94 [1.68]						-0.21 [-0.48]
Anomaly 2		0.35 [0.03]					0.12 [0.15]
Anomaly 3			0.15 [3.57]				-0.12 [-1.10]
Anomaly 4				0.15 [3.54]			0.29 [2.58]
Anomaly 5					-0.97 [-0.86]		-0.13 [-2.78]
Anomaly 6						0.60 [1.26]	0.87 [3.07]
# months	643	648	643	643	643	648	643
$\bar{R}^2(\%)$	2	0	2	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CHI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CHI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.20 [1.99]	0.21 [2.11]	0.21 [2.04]	0.20 [1.96]	0.21 [2.00]	0.22 [2.11]	0.23 [2.22]
Anomaly 1	-4.28 [-1.22]						-1.70 [-0.29]
Anomaly 2		-12.81 [-2.24]					-10.03 [-1.60]
Anomaly 3			-5.98 [-1.66]				-3.43 [-0.48]
Anomaly 4				-4.55 [-1.51]			-1.74 [-0.31]
Anomaly 5					-4.96 [-1.27]		-2.67 [-0.56]
Anomaly 6						-6.43 [-1.51]	-11.70 [-2.41]
mkt	18.77 [6.81]	18.49 [7.30]	18.39 [6.82]	18.25 [6.52]	18.96 [7.12]	21.36 [8.61]	17.30 [5.97]
smb	8.89 [1.80]	8.74 [2.21]	8.07 [1.73]	9.82 [2.36]	9.58 [2.13]	19.67 [3.44]	14.55 [2.40]
hml	8.45 [1.77]	5.95 [1.27]	8.94 [1.87]	8.71 [1.82]	5.05 [1.01]	8.25 [1.76]	8.63 [1.55]
rmw	0.69 [0.12]	8.49 [1.20]	1.04 [0.19]	0.42 [0.08]	-1.65 [-0.34]	-6.82 [-1.31]	5.97 [0.79]
cma	-4.84 [-0.68]	-9.56 [-1.36]	-4.27 [-0.60]	-4.28 [-0.60]	-7.70 [-1.10]	-6.41 [-0.92]	-5.85 [-0.77]
umd	-0.18 [-0.07]	-0.61 [-0.25]	0.24 [0.10]	0.10 [0.04]	-0.14 [-0.06]	-4.60 [-1.35]	-5.61 [-1.62]
# months	680	684	680	680	680	684	680
$\bar{R}^2(\%)$	17	17	17	17	17	17	18

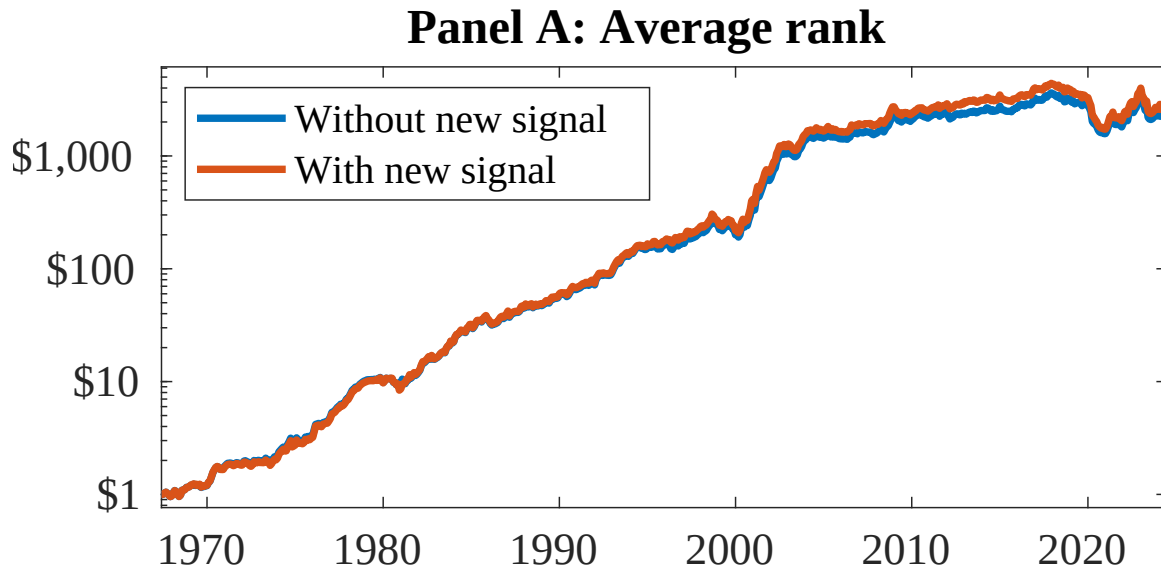


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CHI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Ball, R., Gerakos, J., Linnainmaa, J. T., and Nikolaev, V. (2020). Earnings, retained earnings, and book-to-market in the cross section of expected returns. *Journal of Accounting Research*, 145(1):342–371.
- Bansal, R., Kiku, D., and Yaron, A. (2016). Risks for the long run: Estimation with time aggregation. *Journal of Monetary Economics*, 82:52–69.
- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Bates, T. W., Kahle, K. M., and Stulz, R. M. (2009). Why do u.s. firms hold so much more cash than they used to? *Journal of Finance*, 64(5):1985–2021.
- Boguth, O. and Kuehn, L.-A. (2013). Consumption volatility risk. *Journal of Finance*, 68(6):2589–2615.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

- Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming*.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Jagannathan, R. and Paul, S. (2023). Corporate cash holdings and returns. *Journal of Financial Economics*, 147(2):384–410.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.
- Palazzo, B. and Savor, P. (2024). Cash holdings and the cross-section of stock returns. *Review of Financial Studies*, 37(3):891–930.